Better Than Guessing

Team Members: _____ _____ _____

You can fit a line to anything. The line does not care whether it makes sense. In lecture you built the number that tells you whether a line was worth fitting:

$$R^2 = 1 - \frac{SS_{\text{Residual}}}{SS_{\text{Total}}}$$

Today you will find out where the line stops working — and find a case where R^2 lies to you. Open `Class18_Better_Than_Guessing_Skeleton.ipynb`. Every cell runs as given. Answer here, on paper.

Part 1. The Cricket Line

1.1 Run the first cell and record:

slope _____ **intercept** _____ R^2 _____

State what the slope means in plain English — with units.

1.2 Last class you found $r = 0.80$ between two Pixar rating services and called it a strong correlation. Square it. Does "strong correlation" mean what you assumed it meant?

Part 2. Where the Line Stops Working

2.1 Use your equation to predict the temperature at each chirp rate:

19 chirps/sec → _____ **°F** **40 chirps/sec →** _____ **°F**

And at 0 chirps/sec, the intercept says _____ **°F**

The observed chirp rates run from 14.4 to 20.0. Two of those three predictions are ones you should refuse to make. Which, and what is wrong with each?

Part 3. A Better Number That Is a Worse Model

The notebook now fits a straight line to a second data set. Look at the plot, but do **not** run the R^2 cell yet.

3.1 Predict: will this fit's R^2 be higher or lower than the crickets'? (Circle one.) Now run the cell.

R^2 for the second data set _____ versus the crickets _____

Which data set is the straight line actually the right model for?

3.2 Now compare the two residual plots. Sketch the shape of each, roughly:

crickets	**second data set**

3.3 So: does a high R^2 mean you have the right model? State what the residual plot told you that R^2 could not, in one or two sentences.